

Executing a complete instruction in a computer's Central Processing Unit (CPU) involves a cycle of three fundamental steps: **Fetch**, **Decode**, and **Execute**. This is often referred to as the **Instruction Cycle** or **Fetch-Decode-Execute Cycle**.

Here is an illustration of the execution of a simple instruction: `ADD R1, R2` (Add the contents of Register R2 to Register R1 and store the result in R1).

1 The Instruction Cycle

Stage	Description	Key Components Involved
Fetch	The CPU retrieves the next instruction from the memory location pointed to by the Program Counter (PC) and loads it into the Instruction Register (IR) . The PC is then incremented.	PC, Memory Address Register (MAR), Memory Data Register (MDR), IR.
Decode	The Control Unit (CU) interprets the instruction in the IR to determine the operation to be performed and the addresses of the operands.	IR, Control Unit (CU).
Execute	The Arithmetic Logic Unit (ALU) performs the operation specified by the instruction using the decoded operands, and the result is stored back in the destination (memory or register).	ALU, Registers.

2 Example: Executing `ADD R1, R2`

Assume the following initial state:

- **Program Counter (PC):** Points to memory address `0x100` (where the instruction is stored).
- **Memory at `0x100`:** Contains the instruction `ADD R1, R2`.
- **Register R1:** Contains the value 5.
- **Register R2:** Contains the value 3.

Phase 1: FETCH

1. The CPU copies the address from the **PC** (`0x100`) into the **Memory Address Register (MAR)**.
2. The CPU issues a read command to memory.

3. The instruction **ADD R1, R2** is read from memory and loaded into the **Memory Data Register (MDR)**.
4. The instruction is transferred from the **MDR** to the **Instruction Register (IR)**.
5. The **PC** is incremented to point to the next instruction (0x104, assuming 4-byte instructions).

Phase 2: DECODE

1. The **Control Unit (CU)** reads the instruction **ADD R1, R2** from the **IR**.
2. The CU determines that the operation is **addition (ADD)**.
3. The CU identifies the operands: the contents of **R1** and **R2**.
4. The CU sets up the necessary control signals for the execution phase (e.g., instructing the ALU to perform addition and selecting R1 and R2 as inputs).

Phase 3: EXECUTE

1. The contents of **R1** (5) and **R2** (3) are sent as inputs to the **Arithmetic Logic Unit (ALU)**.
2. The **ALU** performs the addition: $5+3=8$.
3. The result (8) is written back to the destination register, which is **R1**.

Final State

- **Program Counter (PC):** 0x104 (ready for the next instruction).
- **Register R1:** Now contains the new value 8.
- **Register R2:** Still contains the value 3.

The cycle is now complete, and the CPU is ready to start the **Fetch** stage for the instruction at 0x104.